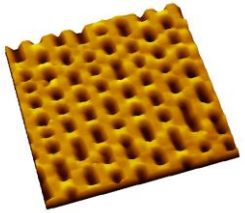
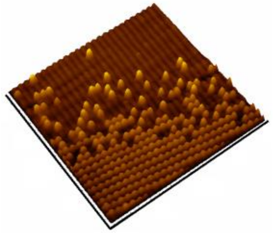
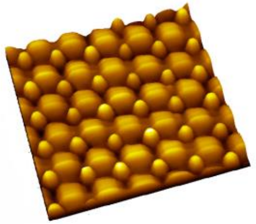
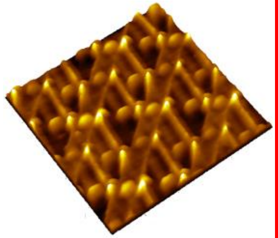
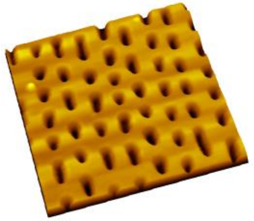
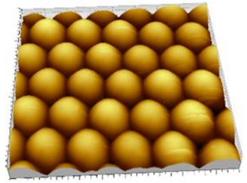

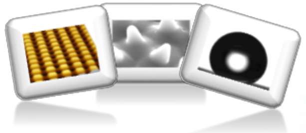

Introduction to Soft Lithography

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Lab web page: https://sites.google.com/site/rmresearchgroup/home

CH62052: Instability
Spring 2022-23

Soft Lithography Methods

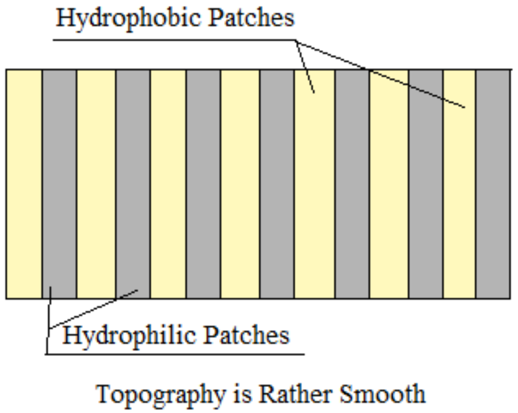
- These are rather specific towards soft surfaces (polymers and gels) as well as for applications which do not require extremely stringent quality control or defect free patterns like micro electronics.
- There are several application areas, where large area meso and nano scale structures are necessary, but even if there are some defects here and there, it is fine.
- Sensors, Bilogical applications, structural color, textured hydrophobicity: progress in all these areas depend on availability of robust, simple, easy to execute and CHEAP patterning techniques that can create large area (~ cm^2) patterns with good reproducibility
- The ability to create defect free structures should be reasonable.

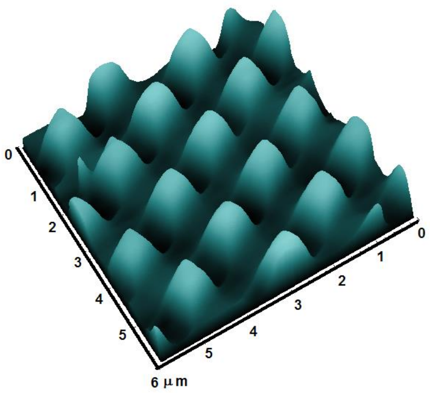
Most cases, the concept works on getting a Perfect Negative Replica of the Original Stamp!

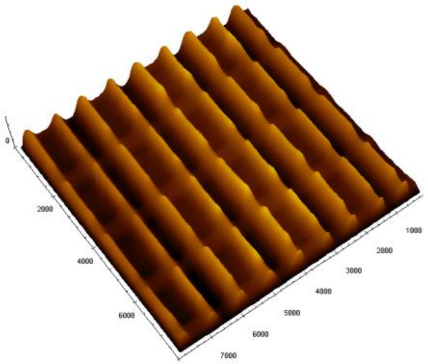
Soft Lithography Techniques: Classifications

Based on the Nature of Patterns:

- Chemical Patterns
- Topographic Patterns



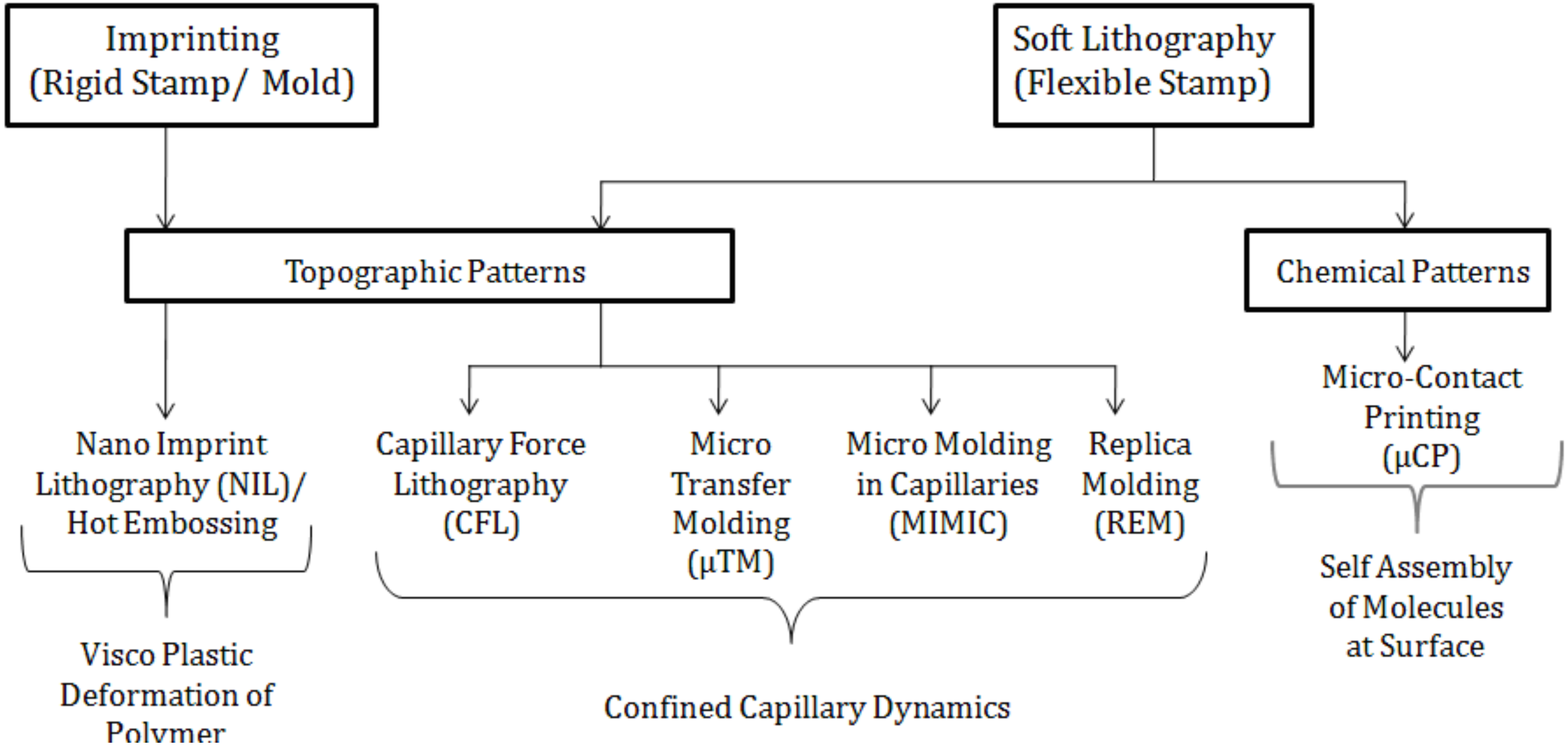




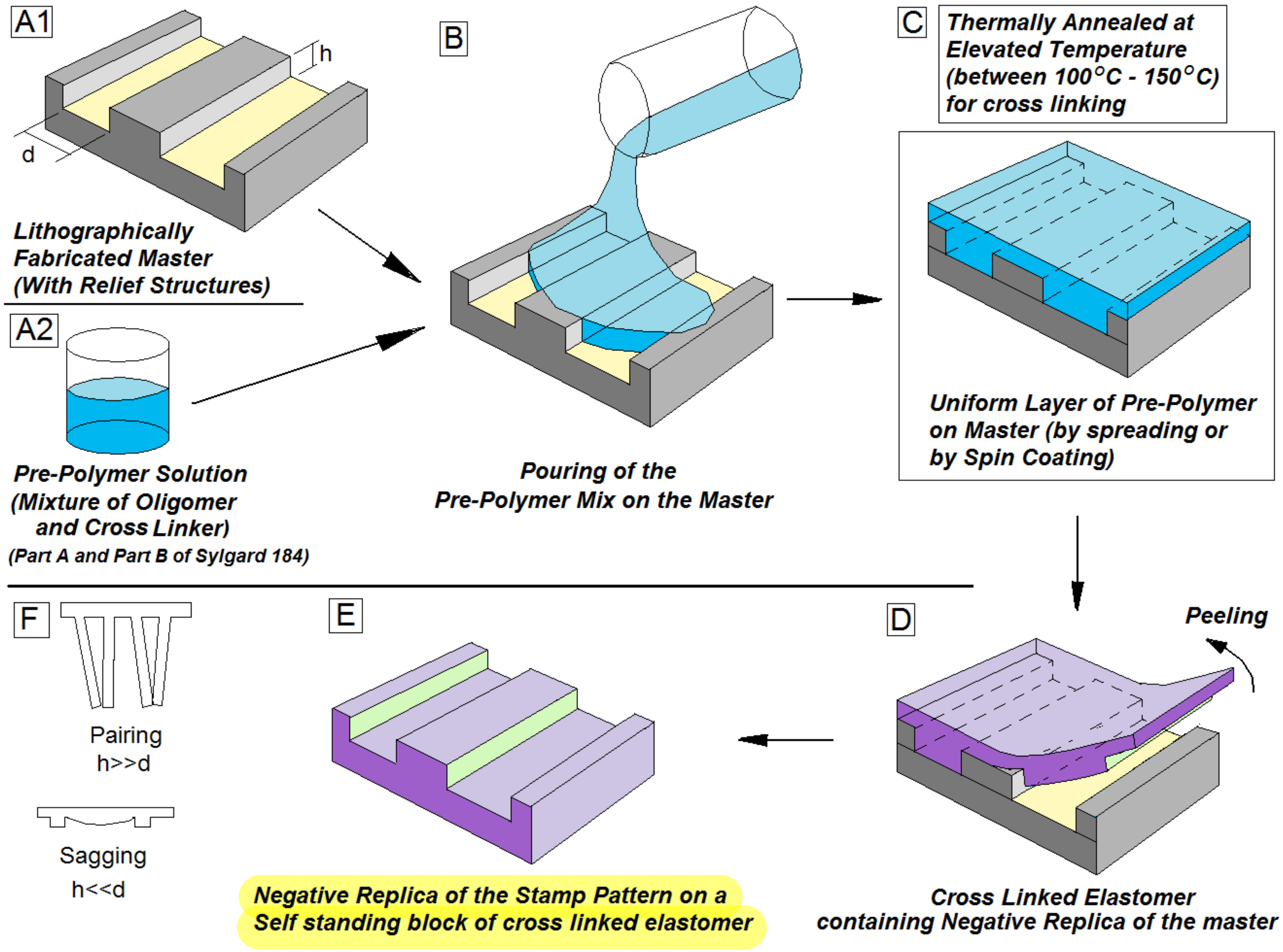
Based on the Nature of Mold or Stamp Used:

- Rigid Stamp
- Flexible Stamp
- Dissolvable Stamp

Soft Lithography Techniques: Classifications



Replica Molding

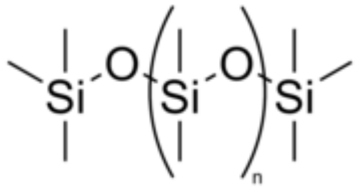


→ most often used to fabricate soft lithography stamp itself

Once it has cross-linked it behaves as a soft solid

Material for Replica Molding

- The material used for REM is Cross linkable Poly-dimethyl siloxane (PDMS), which falls into a general category of materials called elastomers.



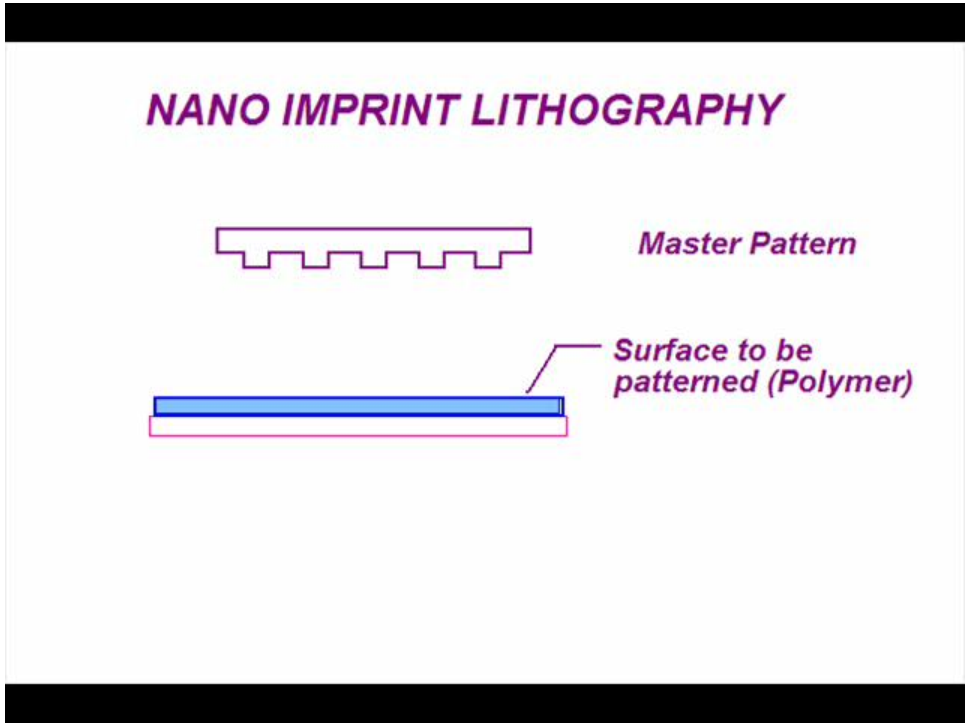
- Elastomers are crosslinked amorphous polymers that are used at temperatures above their glass transition temperature, *Tg.*

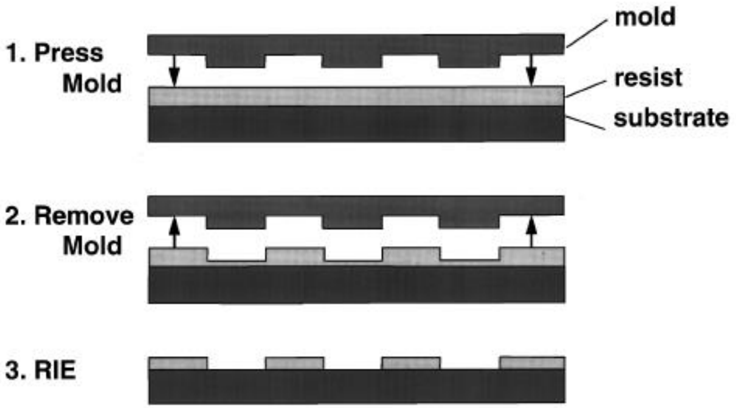
- *Above the glass transition temperature, molecules gain thermal* energy that enables them to move in a coordinated manner, making the elastomers rubbery, soft and flexible.

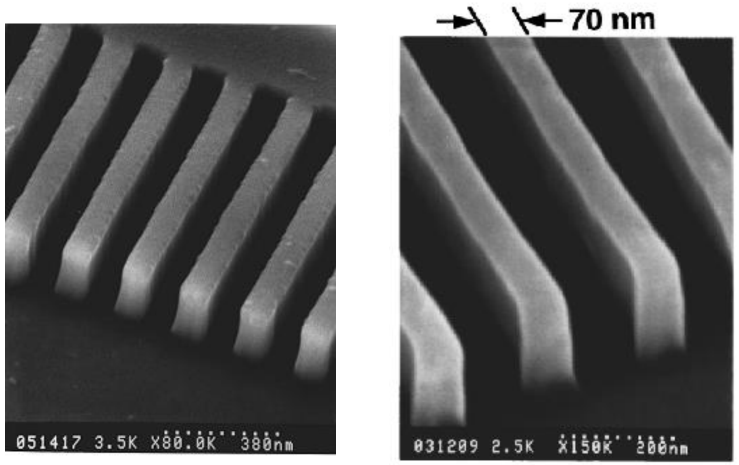
Sylgard Group of Products from Dow – Croning USA

Nano Imprint Lithography (NIL)

J. Vac. Sci. Technol. B, 14, 4129, 1996







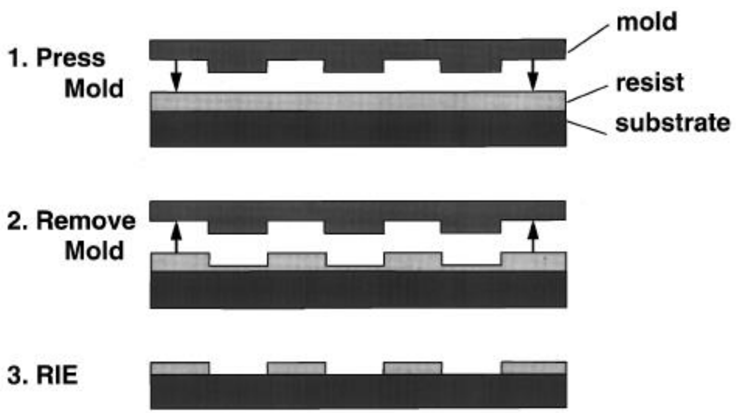
For the original Work

Mold is Rigid:

Patterned Silicon Wafer, Silica or even other materials can be used

Nano Imprint Lithography (NIL)

J. Vac. Sci. Technol. B, 14, 4129, 1996



Advantage:

- Large area patterning capability
- Applicable for many different polymers.
- Resolution achieved ~ 10 nm.
- Possible to achieve patterns over fairly large area.

Some Critical Issues and Limitations

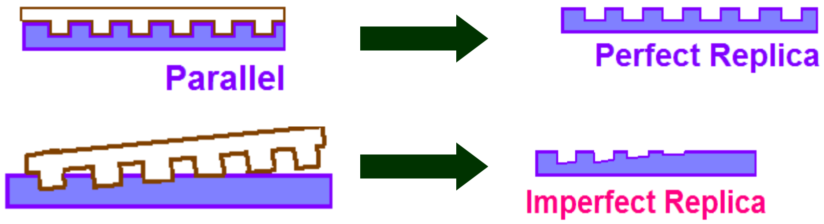
High Temperature

High Pressure

Adhesion of Mold with Resist (Polymer): Severe Chances of Mold Damage

Critical Parallelism between mold and film has to be ensured!

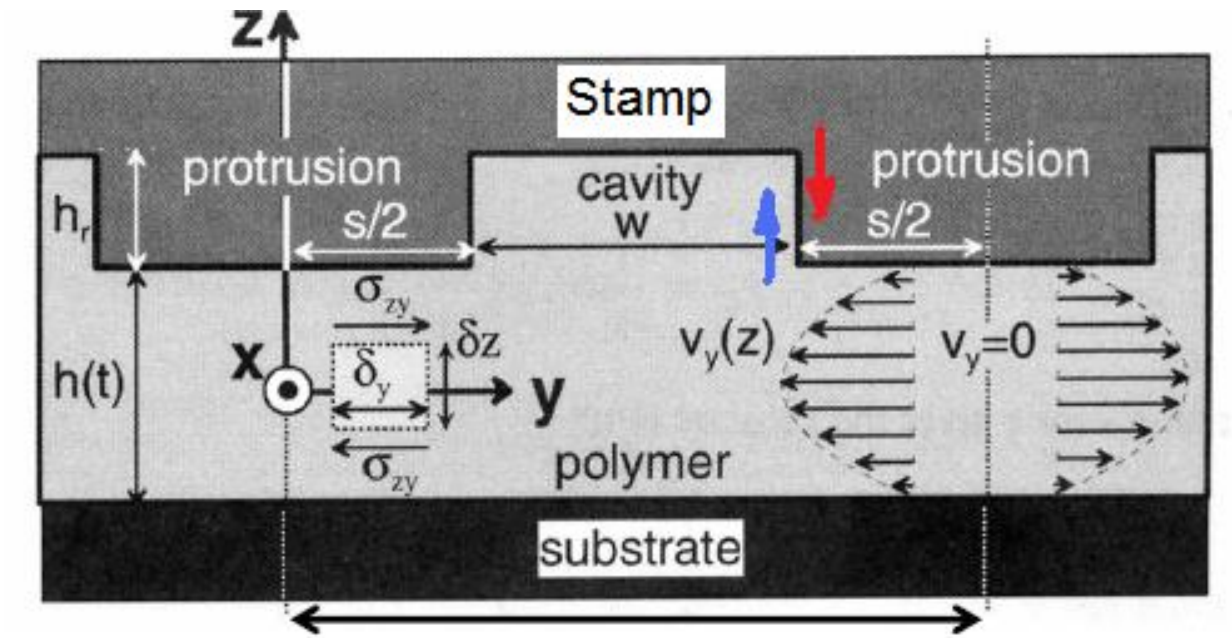
Non Planar Surfaces cannot be patterned

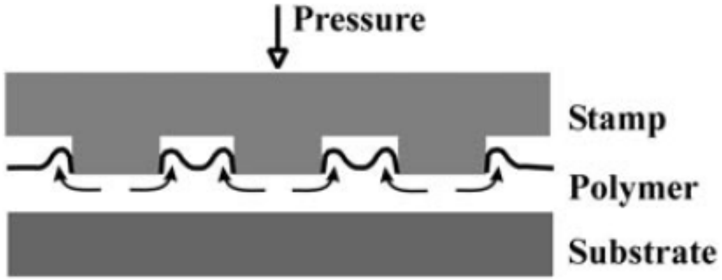


Mold Release agents

Nano Imprint Lithography (NIL)

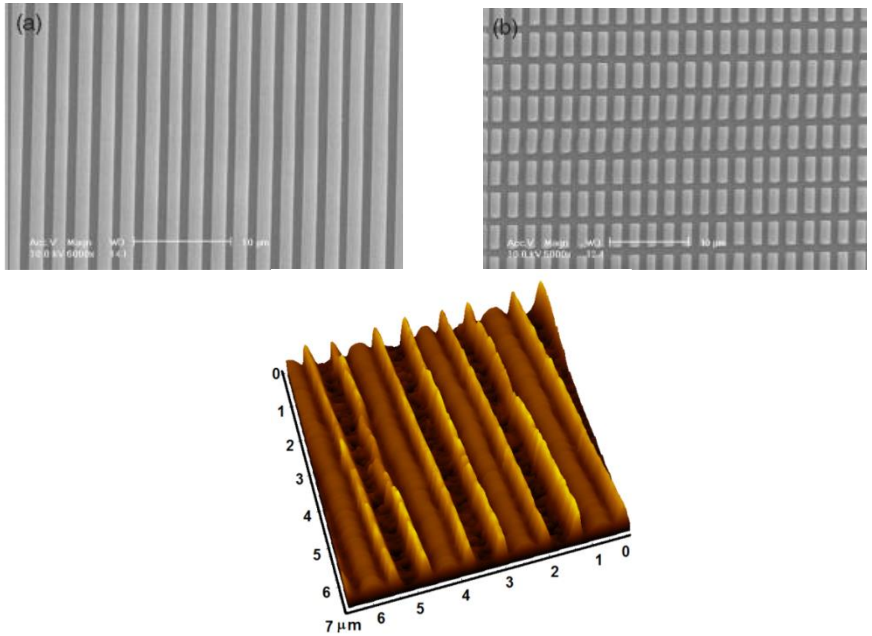
Hydrodynamics and Stresses

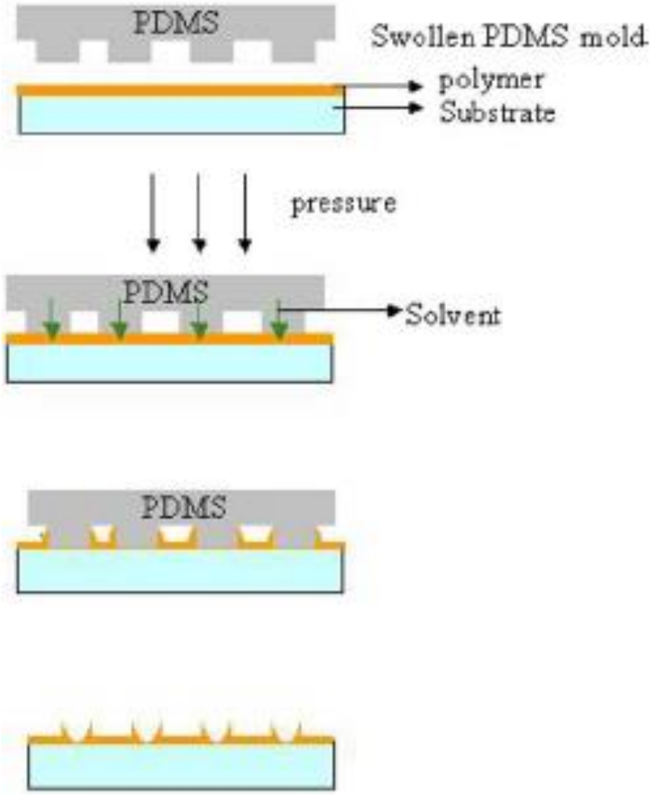




Capillary force Lithography

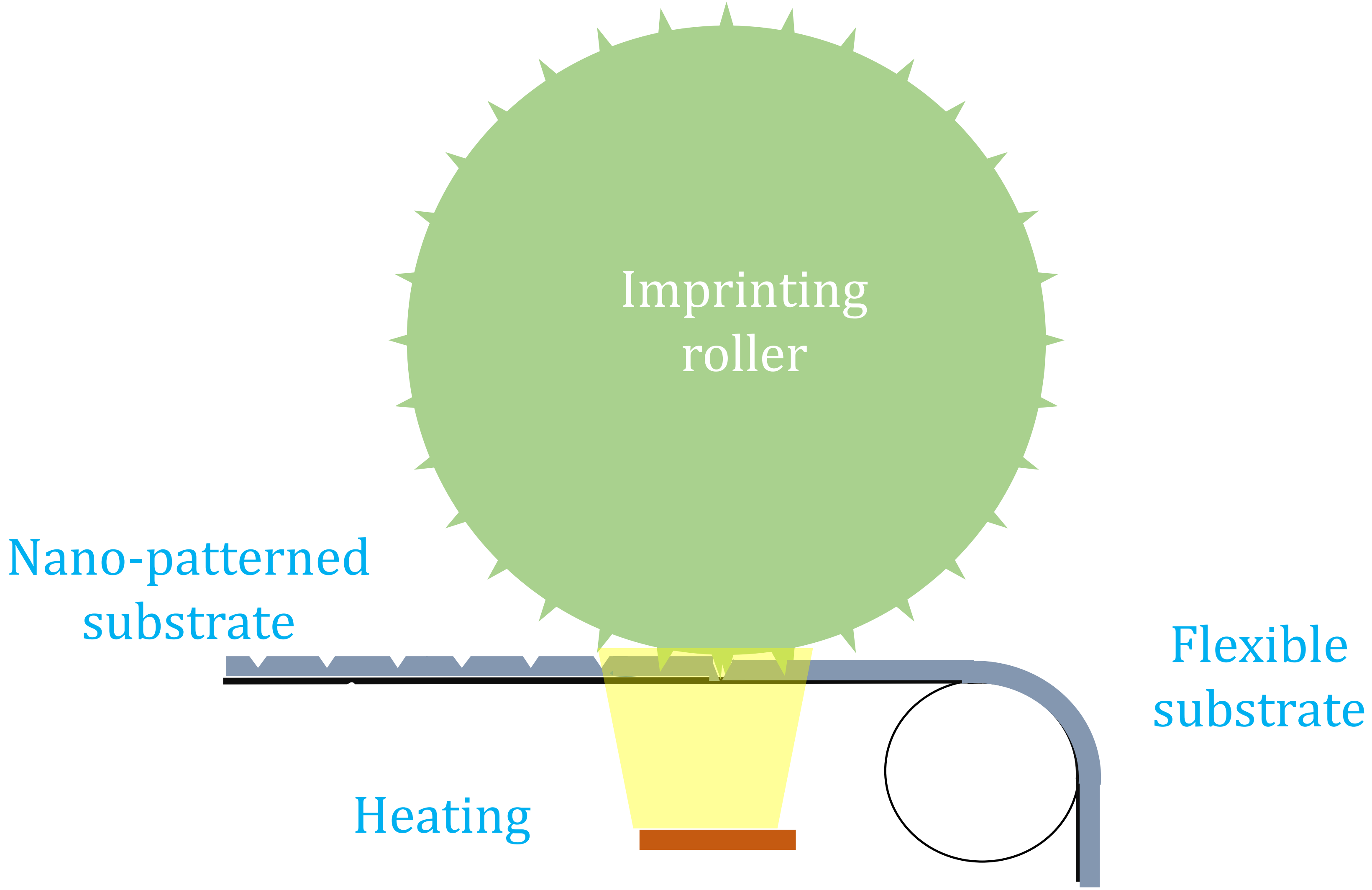
- Heat up the polymer film
- Place mold on top of the mold
- Liquid rises along the walls of the mold.



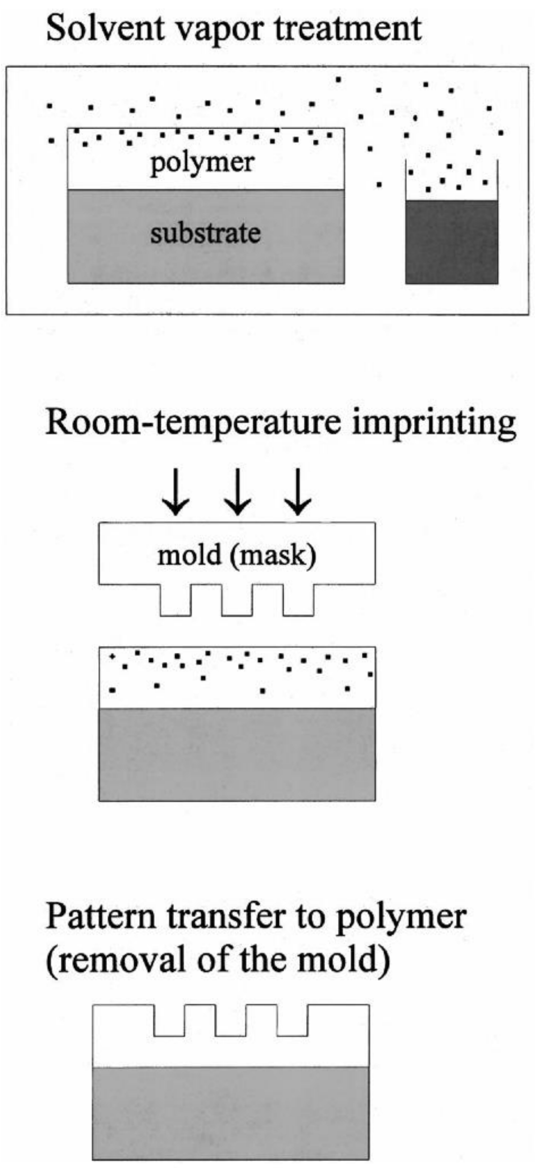


Lee, Adv. Mater, 2001, 13, 1386, Lee, Adv. Mater, 2004, 176, Han, Polymer, 2005, 11099

Roll-to-Roll Nanoimprint Lithography



Solvent Vapor Assisted Nano Imprint Lithography

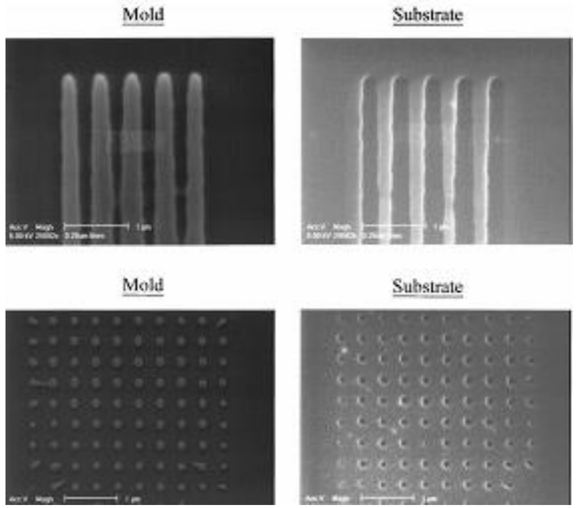


Penetration of solvent molecules into the polymer matrix

Swelling and reduced cohesion

Reduction in Viscosity

Effective reduction in glass transition temperature below room temperature.



Hong Lee: Applied Physics Letters, 76, 870, 2000

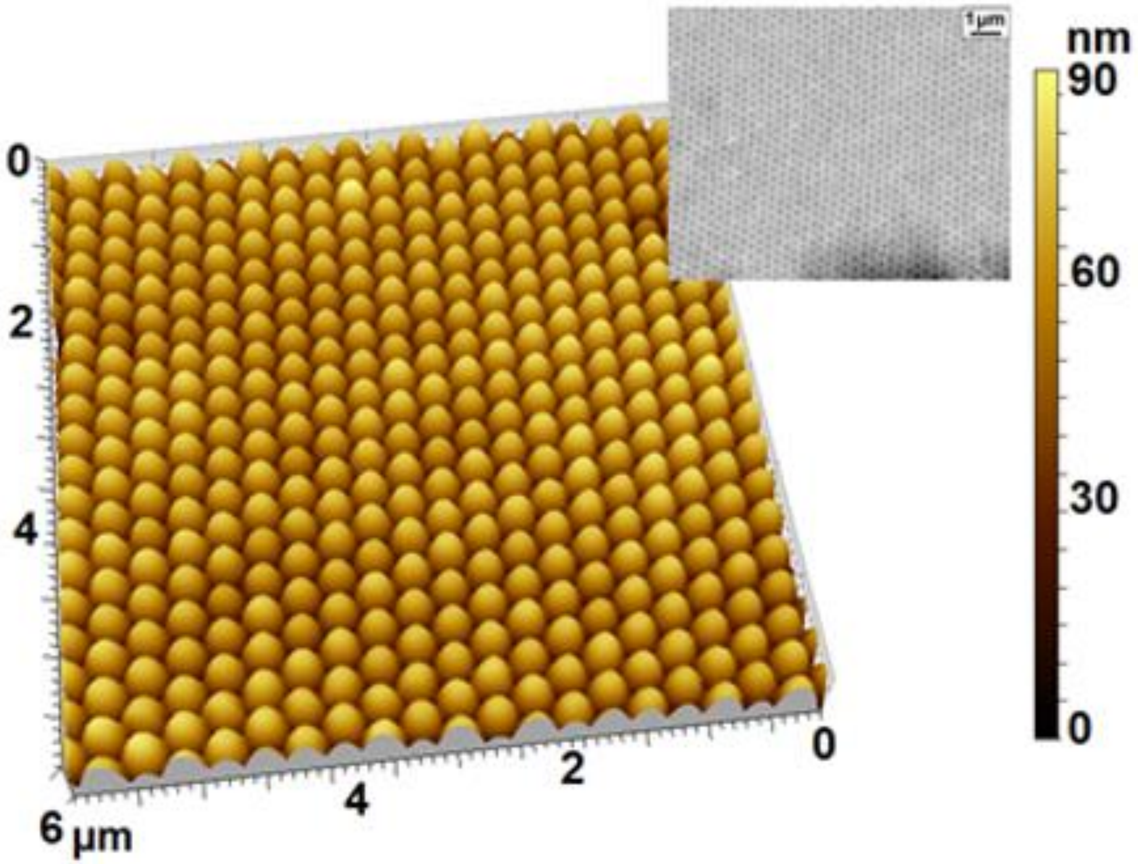
Flory – fox Equation

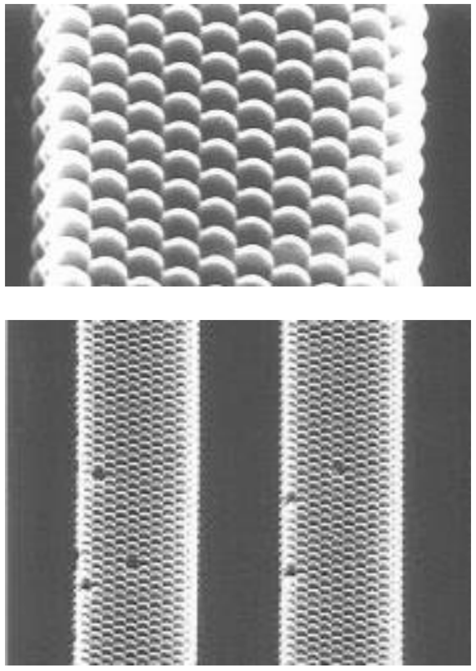
$$\frac{1}{T_G} = \frac{w}{T_G^P} + \frac{1-w}{T_G^S}$$

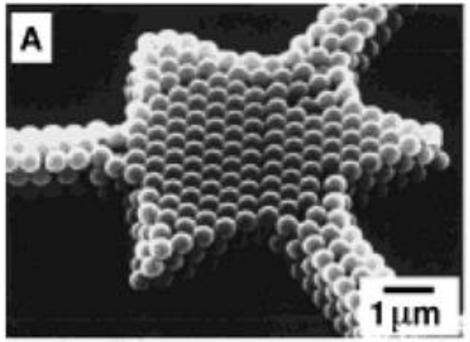
Where T_G^P is the glass transition temperature of the polymer.

T_G^S is the melting temperayure of the solvent

w is the wt. fraction of the polymer in the swollen film







Hexagonal closed pack array of mono dispersed colloidal particles

Can be made by spin coating

Micro molding in Capillaries (MIMIC)

→ In MIMIC you do not start with a film

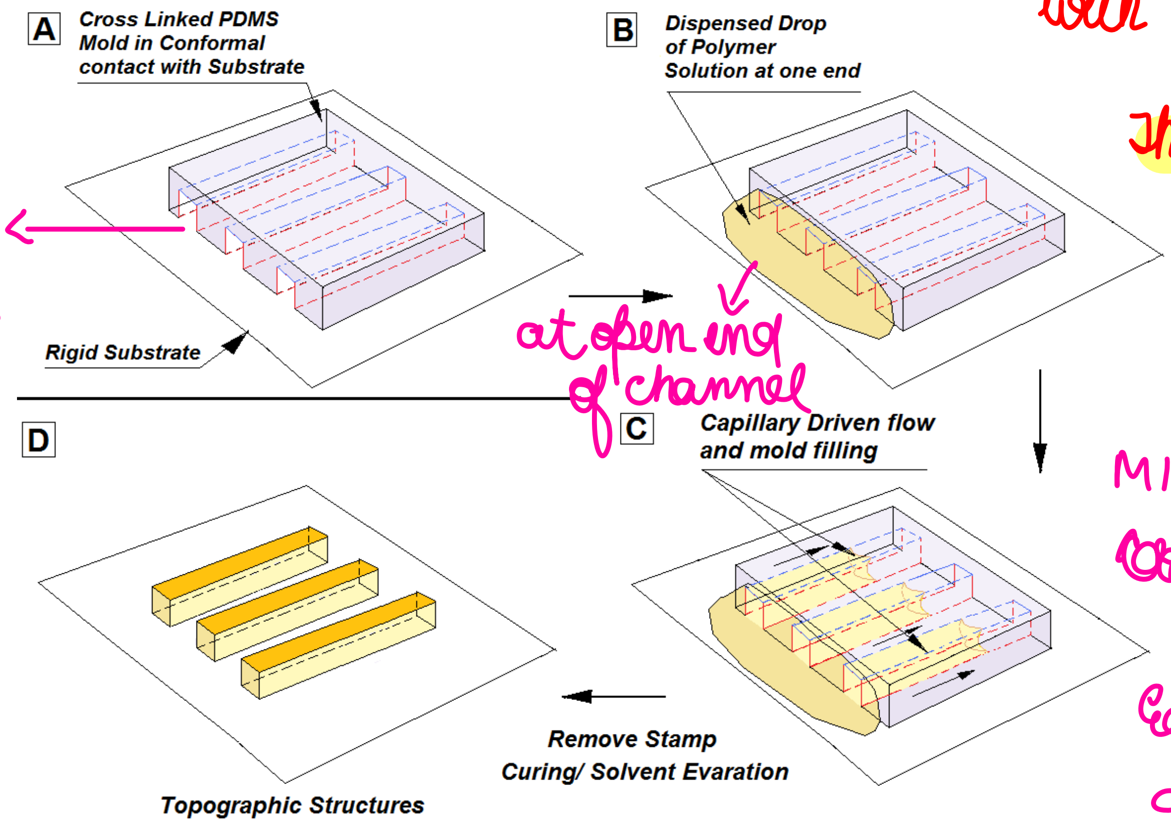
The stamp is bought in conformal contact with BARE substrate

↳ Most Imp **

MIMIC does not require coating of film

Micro channels

at open end of channel



Each channel will act as a capillary

→ will control time for filling of channels (filling rate)

μ = fn(conc)
fn(MW)

- The technique is based on the spontaneous filling of capillaries formed between two surfaces in conformal contact, at least one of which has a recesses relief structure.
- **MIMIC** is a thermodynamically driven process: the liquid fill the capillaries to minimize the free energies of the solid-vapor and the solid-liquid interface.

Liquid may flow in each channel due to capillary

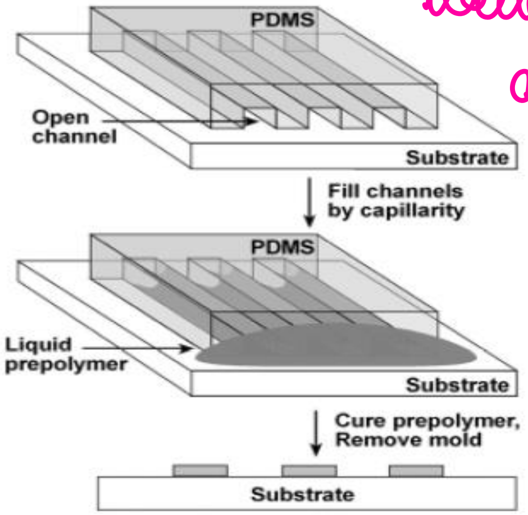
Whitesides, Nature, 376, 1995, 581

So stamp walls have to be preferentially wetted by dispensed polymer soln

Micro molding in Capillaries (MIMIC)

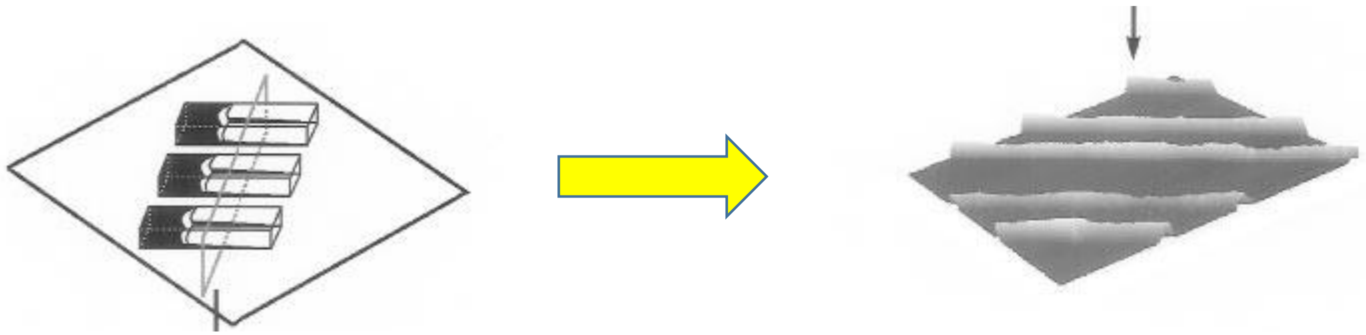
- The filling of capillaries and rate of liquid flow in capillary is determined by the surface tension and viscosity of the liquid.

- Viscosity of liquid is a determining factor that controls the filling rate.



You must have a stamp materials or walls of each of these channels should be preferentially wetted by the polymer solvent

The time allowed for pattern replication is also important as Initially a fluid fills the capillaries only partially, particularly the corner regions.



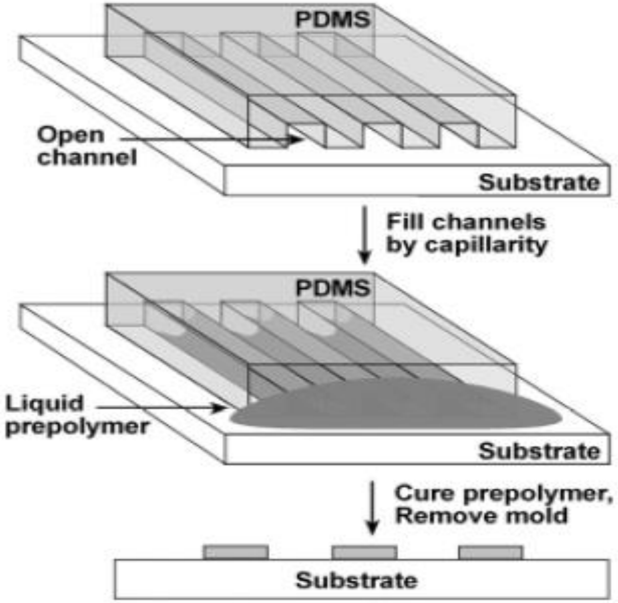
J. AM. CHEM. SOC. 2008, 130, 1177-11

Micro molding in Capillaries (MIMIC)

This equation and all are NOT in the syllabus.

The rate of capillary filling is given by

$$\frac{dz}{dt} = \frac{R\gamma_{LV}\cos\theta}{4\eta z} = \frac{R(\gamma_{SV} - \gamma_{SL})}{4\eta z}$$



η is the kinematic viscosity (μ/ρ)
γ is surface (interfacial) tension of liquid,
R is the radius of the capillary,
z length of the filled section of the capillary

Capillary filling over a short distance (up to 1 cm) can be achieved quickly and efficiently; over a large distance, the rate of filling decreases significantly due to the viscous drag of the fluid in the capillary and the distance over which the fluid has to be transported.

In MIMIC → In order to get a good capillary driven flow you have to ensure that your surface energy of stamp walls is higher, so that you have significant amount of driving force

Micro molding in Capillaries (MIMIC)

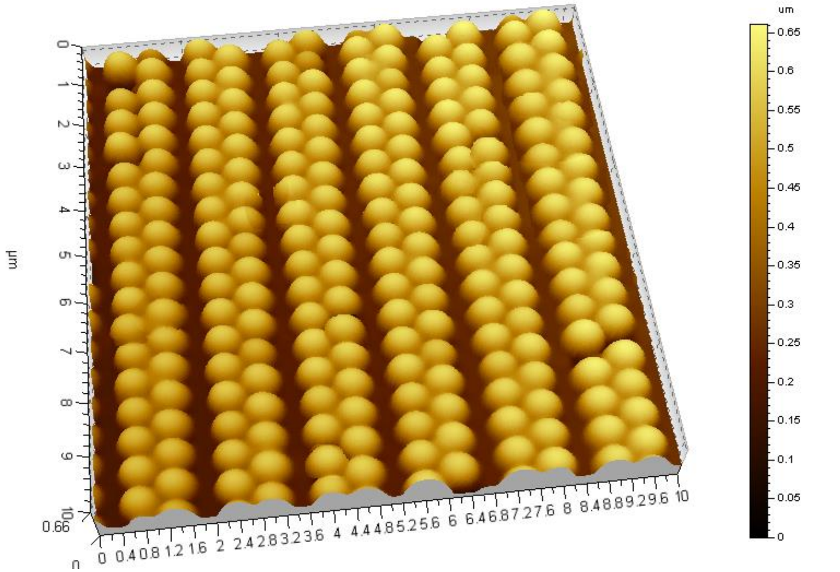
→ Advantage of MIMIC

- The method therefore does not remain limited to a polymer solution only and several functional materials have been patterned by this method.
- In a recent study **Conductive Sub-micrometric Wires of Platinum-Carbonyl Clusters** ($[Pt_{15}(CO)_{18}]^{2-}$) have been patterned by MIMIC
- Ceramic Materials (sol – gel thin films) are also patterned by MIMIC.

driving force is surface tension driven capillarity

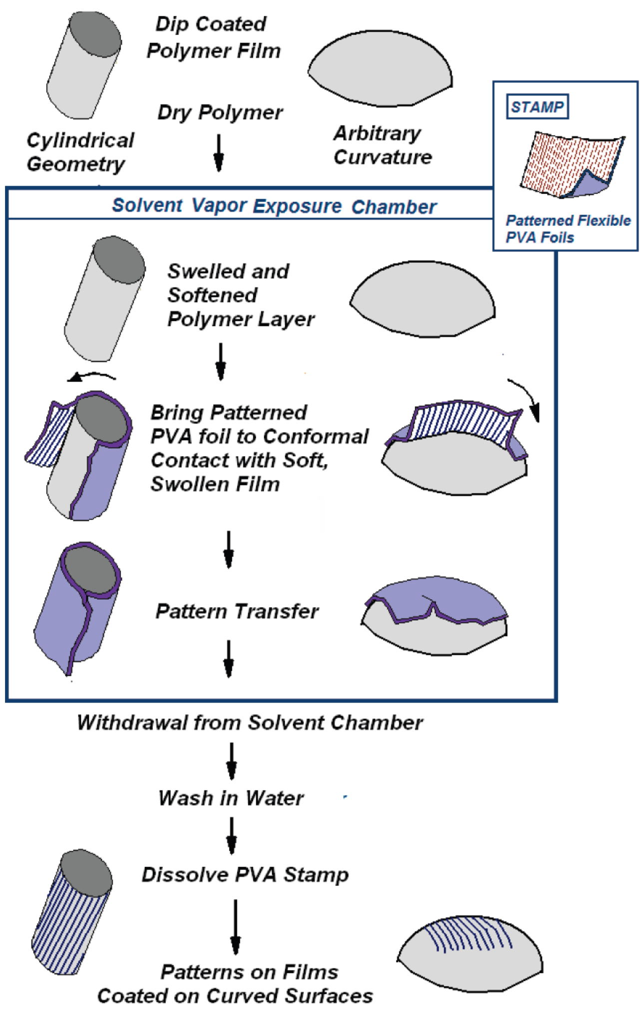
An assembly of Colloids is also possible by MIMIC.

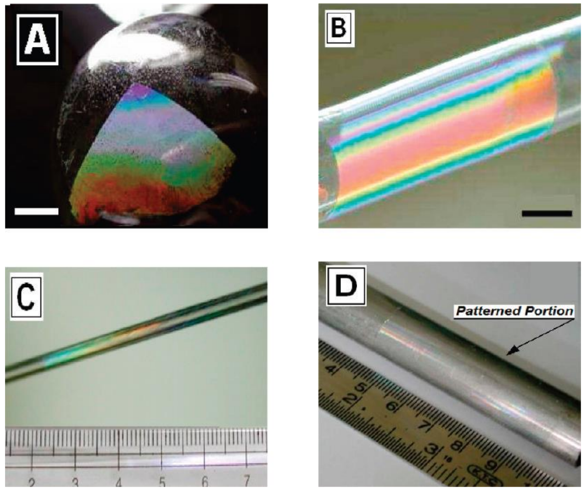
Uniqueness of MIMIC!

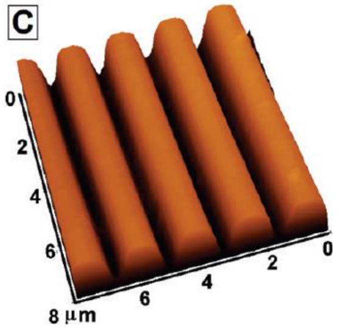


J. Am. Chem. Soc., ***1996, 118 (24), 5722-5731***

Patterning of Curved Surfaces



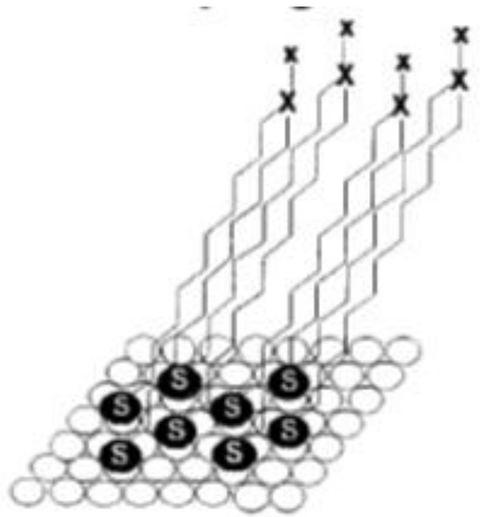




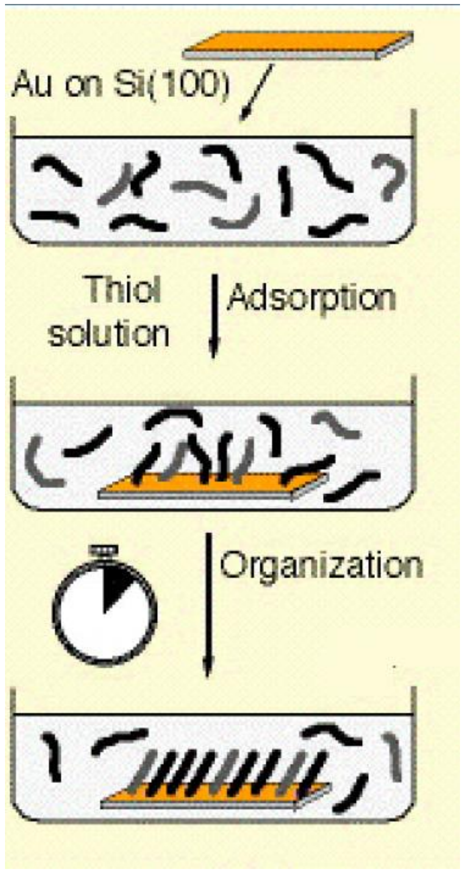
Mukherjee, *Ind. Eng. Chem. Res.* 2009, *48,* 8812–8818

SAM (Self Assembled Monolayer)

- The presence of a ligand ($Y(CH2)_nX$) which is reactive toward the surface ensures the attachment of the thiol molecules with the substrate. The surface properties of the SAM surface (primarily if the SAM coated surface is hydrophobic or hydrophilic) depends on the nature of the head group, X. On the other hand, the binding of the SAM molecules to the surface is determined by the group Y.
- Some surfaces like gold or silver show excellent binding ability towards the thiol molecules such as alkanethiolates.
- .

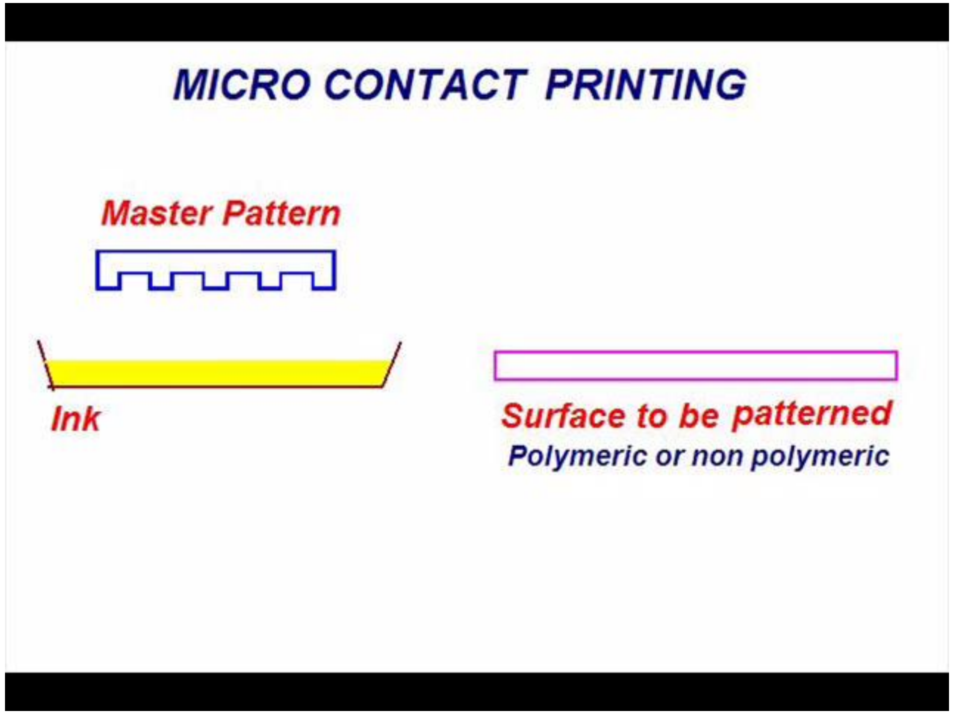


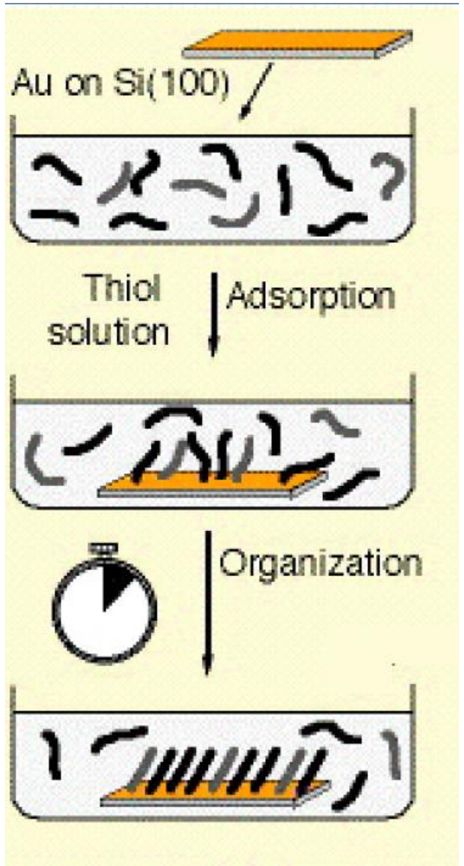
Y is Sulphur in Most cases



Micro Contact Printing (μCP)

- SAM is used as ink (*n-Alkanethiols)*
- *Thiols, Silanes etc.*
- Chemically patterned surfaces obtained.



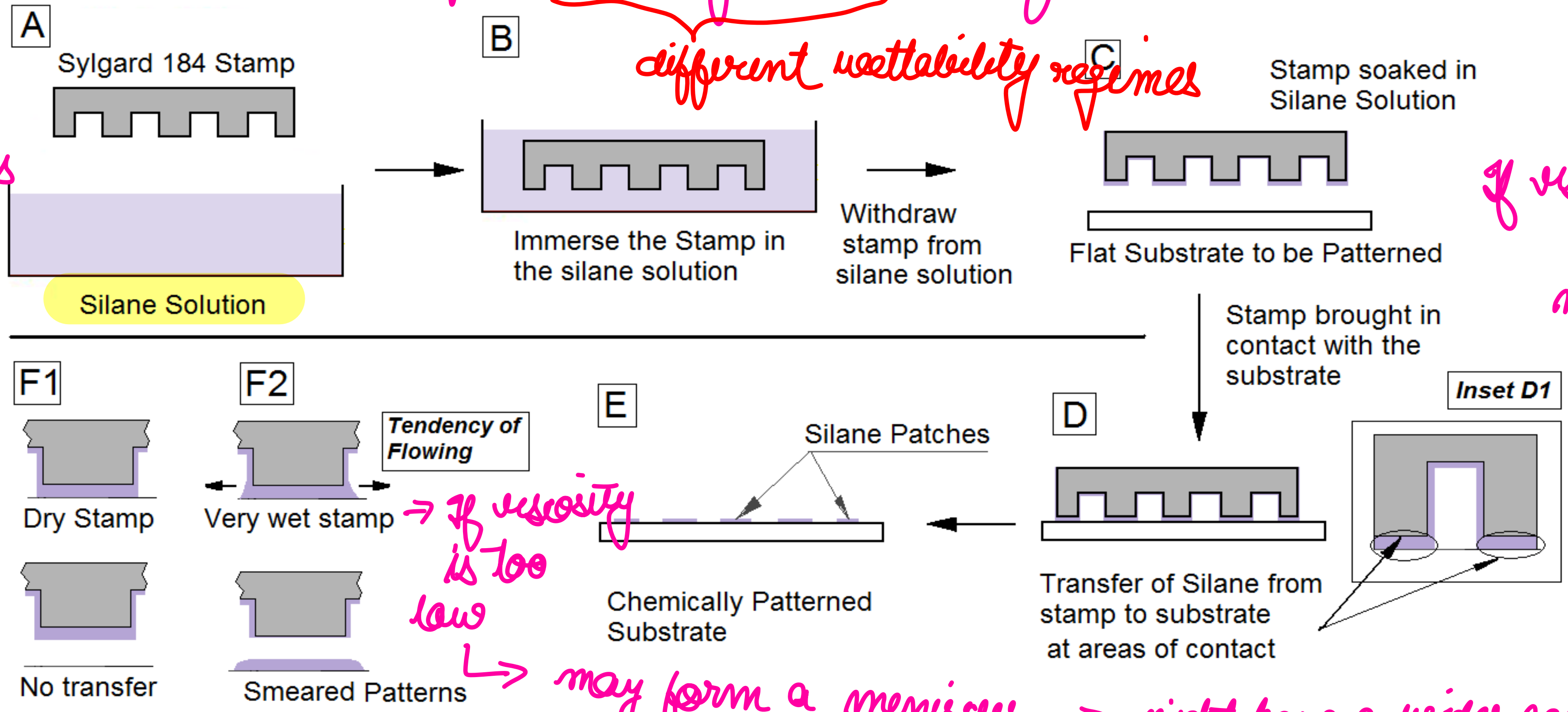


Micro-contact Printing

→ for chemically patterned surface

different wettability regimes

Silane, thiols



If viscosity is too high ↓ molecules do not transfer

→ If viscosity is too low ↳ may form a meniscus → might have a wider zone where molecule transfer

↑ Smearing

- **Possible due to the ability of an elastomeric stamp to conform to a non-planar substrate with minimum distortion of the pattern on its surface.**
- **In this technique a patterned elastomeric stamp (typically PDMS) is inked with an alkanethiol and brought into contact with a gold surface.**
- **A self-assembled mono-layer of alkanethiolates forms at the stamp surface substrate interface.**

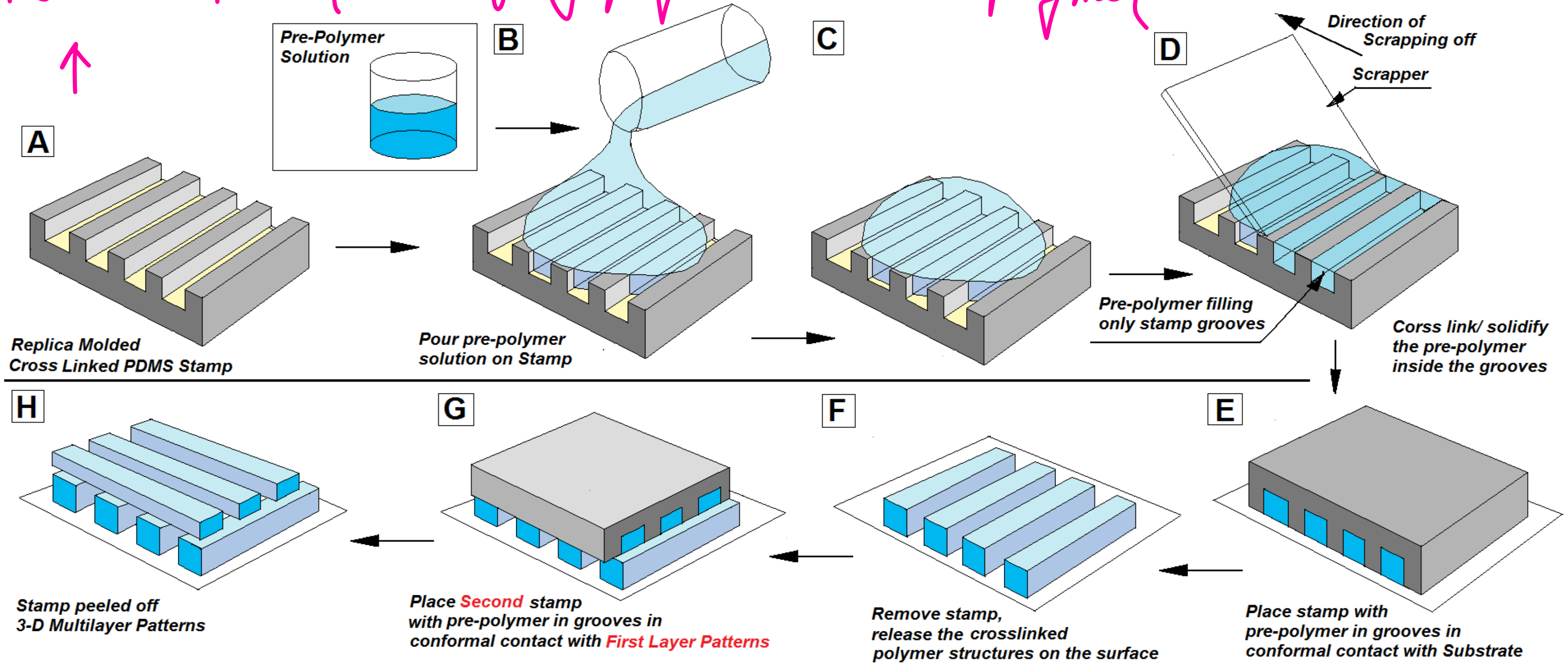
G M Whiteisdes and Amit Kumar, Appl. Phys. Letts., 1993, Science, 1994

Micro Transfer Molding

→ only method that allows you to fabricate 3D structures

PDMS stamp

↳ not fully polymerised → Pre polymer



↳ Stamp is removed by swelling

Multi Layer structures

Soft Lithography Techniques: Classifications

Based on the Pattern Transfer Mechanism:

Chemical Patterns

Are always based on some surface active molecules (Micro Contact Printing) (μCP)

George Whitesides @ Harvard

Topographic Patterns:

Due to **Visco plastic Deformation** of a softened polymer Layer

(Nano Imprint Lithography Group of Techniques) (**NIL**)

Stephen Chou @ Princeton

Or

Due to **Capillary Driven Flow** of a Polymer solution of film in liquid state

(Whitesides, Hong Lee (Korea))

- CFL (Capilary Force Lithography)
- MIMIC (Micro Molding in Capillaries)